

WEEK-2

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Step 1: Create loans Table

The screenshot shows the MySQL Workbench interface. The 'Schemas' pane on the left shows the 'bookflow_db' database with tables 'books', 'members', and 'loans'. The 'SQL' editor in the center contains the following SQL code:

```
1 USE bookflow_db;
2
3 CREATE TABLE loans (
4     loan_id INT PRIMARY KEY,
5     member_id INT,
6     book_id INT,
7     loan_date DATE,
8     FOREIGN KEY (member_id) REFERENCES members(member_id),
9     FOREIGN KEY (book_id) REFERENCES books(book_id)
10 );
11
12
```

The 'Output' pane at the bottom shows the execution results:

#	Time	Action	Message
1	14:57:58	USE bookflow_db	0 row(s) affected
2	14:58:01	CREATE TABLE loans (loan_id INT PRIMARY KEY, member_id INT, book_id INT, loan_date DATE, ...	0 row(s) affected

Step 2: Insert Loan Records

The screenshot shows the MySQL Workbench interface. The 'SQL' editor in the center contains the following SQL code:

```
12 INSERT INTO loans (loan_id, member_id, book_id, loan_date) VALUES
13 (101, 1, 1, '2025-01-05'),
14 (102, 2, 2, '2025-01-10'),
15 (103, 3, 3, '2025-01-20'),
16 (104, 1, 2, '2025-02-01'),
17 (105, 2, 1, '2025-02-05'),
18 (106, 3, 2, '2025-03-01'),
19 (107, 1, 3, '2025-03-07'),
20 (108, 2, 3, '2025-03-15'),
21 (109, 3, 1, '2025-04-01'),
22 (110, 1, 1, '2025-04-15');
```

The 'Result Grid' pane at the bottom shows the inserted data:

loan_id	member_id	book_id	loan_date
102	2	2	2025-01-10
103	3	3	2025-01-20
104	1	2	2025-02-01
105	2	1	2025-02-05
106	3	2	2025-03-01

The 'Output' pane at the bottom shows the execution results:

#	Time	Action	Message
1	14:57:58	USE bookflow_db	0 row(s) affected
2	14:58:01	CREATE TABLE loans (loan_id INT PRIMARY KEY, member_id INT, book_id INT, loan_date DATE, ...	0 row(s) affected
3	15:01:27	INSERT INTO loans (loan_id, member_id, book_id, loan_date) VALUES (101, 1, 1, '2025-01-05'), (102, 2, 2, '2025-01-10'), (103, 3, 3, '2025-01-20'), (104, 1, 2, '2025-02-01'), (105, 2, 1, '2025-02-05'), (106, 3, 2, '2025-03-01'), (107, 1, 3, '2025-03-07'), (108, 2, 3, '2025-03-15'), (109, 3, 1, '2025-04-01'), (110, 1, 1, '2025-04-15');	10 row(s) affected Records: 10 Duplicates: 0 Warnings: 0
4	15:01:36	SELECT * FROM loans LIMIT 0, 1000	10 row(s) returned

Step 3: Display Loans

MySQL Workbench

Local instance MySQL80

File Edit View Query Database Server Tools Scripting Help

Navigator: Schemas

- bookflow_db
 - Tables
 - books
 - members
 - Views
 - Stored Procedures
 - Functions
- sys

Administration Schemas

Information

No object selected

library sql_commands week1 week2*

Limit to 1000 rows

```
25
26 • SELECT
27     m.full_name AS Member_Name,
28     b.title AS Book_Title,
29     l.loan_date
30 FROM loans l
31 INNER JOIN members m
32 ON l.member_id = m.member_id
```

Result Grid

Member_Name	Book_Title	loan_date
Mohammad Rehan	The Alchemist	2025-01-05
Mohammad Rehan	Clean Code	2025-02-01
Mohammad Rehan	Atomic Habits	2025-03-07
Mohammad Rehan	The Alchemist	2025-04-15
Rohit BJ	Clean Code	2025-01-10
Rohit BJ	The Alchemist	2025-02-05
Rohit BJ	Atomic Habits	2025-03-15
Ravi Verma	Atomic Habits	2025-01-20
Ravi Verma	Clean Code	2025-03-01
Ravi Verma	The Alchemist	2025-04-01

Result 3

Output

Action Output

#	Time	Action	Message
1	14:57:58	USE bookflow_db	0 row(s) affected
2	14:58:01	CREATE TABLE loans (loan_id INT PRIMARY KEY, member_id INT, book_id INT, loan_date DATE, ...	0 row(s) affected
3	15:01:27	INSERT INTO loans (loan_id, member_id, book_id, loan_date) VALUES (101, 1, 1, '2025-01-05'), (102, 2, 2, '202...	10 row(s) affected Rec...
4	15:01:36	SELECT * FROM loans LIMIT 0, 1000	10 row(s) returned
5	15:03:39	SELECT * FROM loans LIMIT 0, 1000	10 row(s) returned
6	15:04:03	SELECT m.full_name AS Member_Name, b.title AS Book_Title, l.loan_date FROM loans l INNER JOIN m...	10 row(s) returned

Step 4: INNER JOIN Query

MySQL Workbench

Local instance MySQL80

File Edit View Query Database Server Tools Scripting Help

Navigator: Schemas

- bookflow_db
 - Tables
 - books
 - members
 - Views
 - Stored Procedures
 - Functions
- sys

Administration Schemas

Information

No object selected

library sql_commands week1 week2*

Limit to 1000 rows

```
23
24 • SELECT * FROM loans;
25
26
27
```

Result Grid

loan_id	member_id	book_id	loan_date
101	1	1	2025-01-05
102	2	2	2025-01-10
103	3	3	2025-01-20
104	1	2	2025-02-01
105	2	1	2025-02-05
106	3	2	2025-03-01
107	1	3	2025-03-07
108	2	3	2025-03-15
109	3	1	2025-04-01
110	1	1	2025-04-15

loans 2

Output

Action Output

#	Time	Action	Message
1	14:57:58	USE bookflow_db	0 row(s) affected
2	14:58:01	CREATE TABLE loans (loan_id INT PRIMARY KEY, member_id INT, book_id INT, loan_date DATE, ...	0 row(s) affected
3	15:01:27	INSERT INTO loans (loan_id, member_id, book_id, loan_date) VALUES (101, 1, 1, '2025-01-05'), (102, 2, 2, '202...	10 row(s) affected Rec...
4	15:01:36	SELECT * FROM loans LIMIT 0, 1000	10 row(s) returned
5	15:03:39	SELECT * FROM loans LIMIT 0, 1000	10 row(s) returned

Step 5: GROUP BY Query

The screenshot shows the MySQL Workbench interface. The left sidebar displays the 'bookflow_db' database with tables 'books' and 'members'. The main editor window contains the following SQL query:

```
35
36 * SELECT
37     published_year,
38     COUNT(book_id) AS Total_Books
39 FROM books
40 GROUP BY published_year
41 ORDER BY published_year;
42
```

The 'Result Grid' shows the output of the query:

published_year	Total_Books
1988	1
2008	1
2018	1

The bottom panel shows the 'Action Output' with the following messages:

#	Time	Action	Message
2	14:58:01	CREATE TABLE loans (loan_id INT PRIMARY KEY, member_id INT, book_id INT, loan_date DAT...	0 row(s) affected
3	15:01:27	INSERT INTO loans (loan_id, member_id, book_id, loan_date) VALUES (101, 1, 1, 2025-01-05), (102, 2, 2, '2...	10 row(s) affected Rec
4	15:01:36	SELECT * FROM loans LIMIT 0, 1000	10 row(s) returned
5	15:03:39	SELECT * FROM loans LIMIT 0, 1000	10 row(s) returned
6	15:04:03	SELECT m.full_name AS Member_Name, b.title AS Book_Title, l.loan_date FROM loans l INNER JOIN ...	10 row(s) returned
7	15:06:00	SELECT published_year, COUNT(book_id) AS Total_Books FROM books GROUP BY published_year O...	3 row(s) returned

Step 6: Create donation_history Table

The screenshot shows the MySQL Workbench interface. The left sidebar displays the 'bookflow_db' database with tables 'books' and 'members'. The main editor window contains the following SQL query:

```
30 FROM loans l
31 INNER JOIN members m
32 ON l.member_id = m.member_id
33 INNER JOIN books b
34 ON l.book_id = b.book_id;
35
36 * SELECT
37     published_year,
38     COUNT(book_id) AS Total_Books
39 FROM books
40 GROUP BY published_year
41 ORDER BY published_year;
42 * CREATE TABLE donation_history (
43     donation_id INT PRIMARY KEY,
44     book_id INT,
45     donor_name VARCHAR(100),
46     donation_date DATE,
47     FOREIGN KEY (book_id) REFERENCES books(book_id)
48 );
49
```

The 'Output' panel shows the 'Action Output' with the following messages:

#	Time	Action	Message
4	15:01:36	SELECT * FROM loans LIMIT 0, 1000	10 row(s) returned
5	15:03:39	SELECT * FROM loans LIMIT 0, 1000	10 row(s) returned
6	15:04:03	SELECT m.full_name AS Member_Name, b.title AS Book_Title, l.loan_date FROM loans l INNER JOIN ...	10 row(s) returned
7	15:06:00	SELECT published_year, COUNT(book_id) AS Total_Books FROM books GROUP BY published_year O...	3 row(s) returned
8	15:06:31	SELECT published_year, COUNT(book_id) AS Total_Books FROM books GROUP BY published_year O...	3 row(s) returned
9	15:06:38	CREATE TABLE donation_history (donation_id INT PRIMARY KEY, book_id INT, donor_name VARC...	0 row(s) affected

Step 7: Insert Donation Record

MySQL Workbench

Local instance MySQL80 x

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Navigator: library sql_commands week1 week2*

SCHEMAS

Filter objects

bookflow_db

- Tables
 - books
 - members
- Views
- Stored Procedures
- Functions

sys

Administration Schemas

Information

No object selected

```
35
36 SELECT
37     published_year,
38     COUNT(book_id) AS Total_Books
39 FROM books
40 GROUP BY published_year
41 ORDER BY published_year;
42 CREATE TABLE donation_history (
43     donation_id INT PRIMARY KEY,
44     book_id INT,
45     donor_name VARCHAR(100),
46     donation_date DATE,
47     FOREIGN KEY (book_id) REFERENCES books(book_id)
48 );
49
50 INSERT INTO donation_history
51 (donation_id, book_id, donor_name, donation_date)
52 VALUES
53 (1, 1, 'Mohammad Rehan', '2026-08-04');
54
```

Output

Action Output

#	Time	Action	Message
5	15:03:39	SELECT * FROM loans LIMIT 0, 1000	10 row(s) returned
6	15:04:03	SELECT m.full_name AS Member_Name, b.title AS Book_Title, l.loan_date FROM loans l INNER JOIN ...	10 row(s) returned
7	15:06:00	SELECT published_year, COUNT(book_id) AS Total_Books FROM books GROUP BY published_year O...	3 row(s) returned
8	15:06:31	SELECT published_year, COUNT(book_id) AS Total_Books FROM books GROUP BY published_year O...	3 row(s) returned
9	15:06:38	CREATE TABLE donation_history (donation_id INT PRIMARY KEY, book_id INT, donor_name VARC...	0 row(s) affected
10	15:07:36	INSERT INTO donation_history (donation_id, book_id, donor_name, donation_date) VALUES (1, 1, 'Mohamma...	1 row(s) affected

Object Info Session

Step 8: Display Donation History

MySQL Workbench

Local instance MySQL80 x

File Edit View Query Database Server Tools Scripting Help

Navigator: library sql_commands week1 week2*

SCHEMAS

Filter objects

bookflow_db

- Tables
 - books
 - members
- Views
- Stored Procedures
- Functions

sys

Administration Schemas

Information

No object selected

```
49
50 INSERT INTO donation_history
51 (donation_id, book_id, donor_name, donation_date)
52 VALUES
53 (1, 1, 'Mohammad Rehan', '2026-08-04');
54
55 SELECT * FROM donation_history;
56
```

Result Grid

donation_id	book_id	donor_name	donation_date
1	1	Mohammad Rehan	2026-08-04

donation_history6 x

Output

Action Output

#	Time	Action	Message
6	15:04:03	SELECT m.full_name AS Member_Name, b.title AS Book_Title, l.loan_date FROM loans l INNER JOIN ...	10 row(s) returned
7	15:06:00	SELECT published_year, COUNT(book_id) AS Total_Books FROM books GROUP BY published_year O...	3 row(s) returned
8	15:06:31	SELECT published_year, COUNT(book_id) AS Total_Books FROM books GROUP BY published_year O...	3 row(s) returned
9	15:06:38	CREATE TABLE donation_history (donation_id INT PRIMARY KEY, book_id INT, donor_name VARC...	0 row(s) affected
10	15:07:36	INSERT INTO donation_history (donation_id, book_id, donor_name, donation_date) VALUES (1, 1, 'Mohamma...	1 row(s) affected
11	15:08:00	SELECT * FROM donation_history LIMIT 0, 1000	1 row(s) returned

Object Info Session